

```
#importing Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
#load dataset
df=pd.read_csv('/content/healthcare-dataset-stroke-data.csv')
#display first 10 rows
print("First 10 rows of dataset:- ")
print(df.head(10))
```

First 10 rows of dataset:-

	id	gender	age	hypertension	heart_disease	ever_married	\
0	9046	Male	67.0	0	1	Yes	
1	51676	Female	61.0	0	0	Yes	
2	31112	Male	80.0	0	1	Yes	
3	60182	Female	49.0	0	0	Yes	
4	1665	Female	79.0	1	0	Yes	
5	56669	Male	81.0	0	0	Yes	
6	53882	Male	74.0	1	1	Yes	
7	10434	Female	69.0	0	0	No	
8	27419	Female	59.0	0	0	Yes	
9	60491	Female	78.0	0	0	Yes	

	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	\
0	Private	Urban	228.69	36.6	formerly smoked	
1	Self-employed	Rural	202.21	NaN	never smoked	
2	Private	Rural	105.92	32.5	never smoked	
3	Private	Urban	171.23	34.4	smokes	
4	Self-employed	Rural	174.12	24.0	never smoked	
5	Private	Urban	186.21	29.0	formerly smoked	
6	Private	Rural	70.09	27.4	never smoked	
7	Private	Urban	94.39	22.8	never smoked	
8	Private	Rural	76.15	NaN	Unknown	
9	Private	Urban	58.57	24.2	Unknown	

stroke

0	1
1	1
2	1
3	1
4	1
5	1
6	1
7	1
8	1
9	1

```
#Basic info
print("\nBasic Info:-")
print(df.info())
```

Basic Info:-

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 5110 entries, 0 to 5109
```

```
Data columns (total 12 columns):
```

#	Column	Non-Null Count	Dtype
0	id	5110 non-null	int64
1	gender	5110 non-null	object
2	age	5110 non-null	float64
3	hypertension	5110 non-null	int64
4	heart_disease	5110 non-null	int64
5	ever_married	5110 non-null	object
6	work_type	5110 non-null	object
7	Residence_type	5110 non-null	object
8	avg_glucose_level	5110 non-null	float64
9	bmi	4909 non-null	float64
10	smoking_status	5110 non-null	object
11	stroke	5110 non-null	int64

```
dtypes: float64(3), int64(4), object(5)
```

```
memory usage: 479.2+ KB
```

```
None
```

```
#check for missing values
print("\nMissing values:-")
print(df.isnull().sum())
```

Missing values:-

id	0
gender	0

```

age            0
hypertension   0
heart_disease  0
ever_married   0
work_type      0
Residence_type 0
avg_glucose_level 0
bmi            201
smoking_status 0
stroke         0
dtype: int64

```

```

#drop duplicates
df= df.drop_duplicates()

```

```

#Basic Statistics
print("\nBasic Statistics:-")
print(df.describe())

```

```

Basic Statistics:-

```

	id	age	hypertension	heart_disease \
count	5110.000000	5110.000000	5110.000000	5110.000000
mean	36517.829354	43.226614	0.097456	0.054012
std	21161.721625	22.612647	0.296607	0.226063
min	67.000000	0.080000	0.000000	0.000000
25%	17741.250000	25.000000	0.000000	0.000000
50%	36932.000000	45.000000	0.000000	0.000000
75%	54682.000000	61.000000	0.000000	0.000000
max	72940.000000	82.000000	1.000000	1.000000

	avg_glucose_level	bmi	stroke
count	5110.000000	4909.000000	5110.000000
mean	106.147677	28.893237	0.048728
std	45.283560	7.854067	0.215320
min	55.120000	10.300000	0.000000
25%	77.245000	23.500000	0.000000
50%	91.885000	28.100000	0.000000
75%	114.090000	33.100000	0.000000
max	271.740000	97.600000	1.000000

```

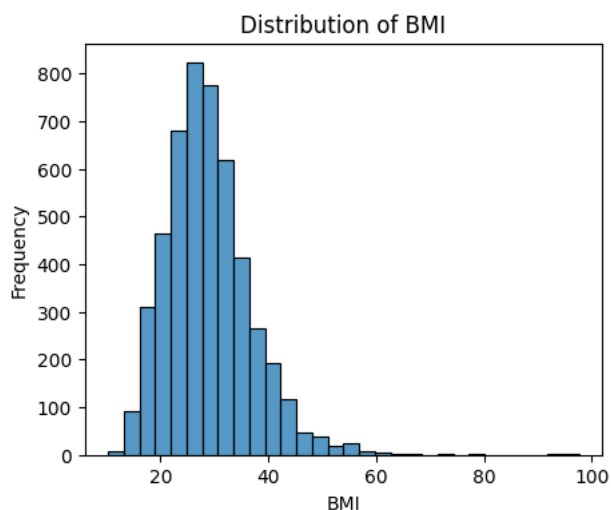
#data Cleaning
df.columns = df.columns.str.strip()
df = df.dropna()
#correlation for numeric features
corr = df.corr(numeric_only=True)

```

```

#Data visulazation
#Distribution of BMI(Body Mass Index)
plt.figure(figsize=(5,4))
sns.histplot(df['bmi'], bins=30)
plt.title('Distribution of BMI')
plt.xlabel('BMI')
plt.ylabel('Frequency')
plt.show()

```

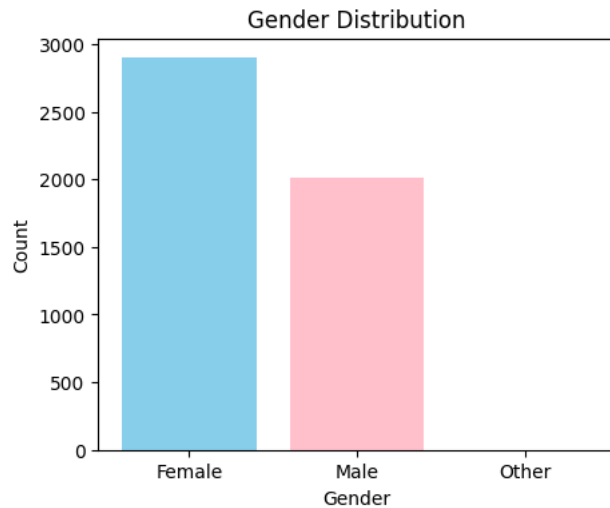


```

#Bar Chart for Gender
gender_count= df['gender'].value_counts()
plt.figure(figsize=(5,4))
plt.bar(gender_count.index, gender_count.values,color=['skyblue','pink'])

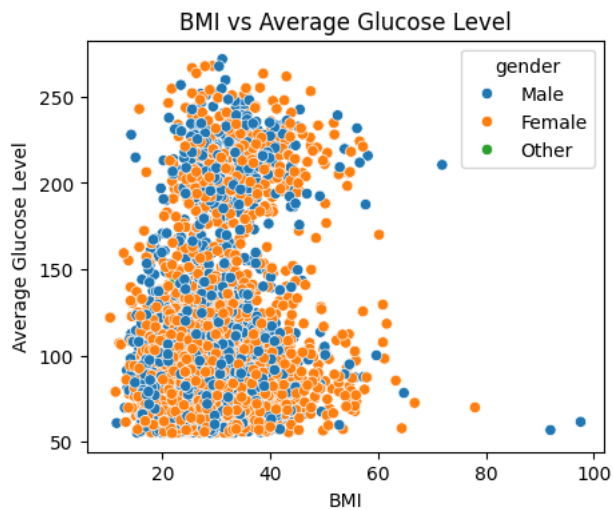
```

```
plt.title('Gender Distribution')
plt.xlabel('Gender')
plt.ylabel('Count')
plt.show()
```

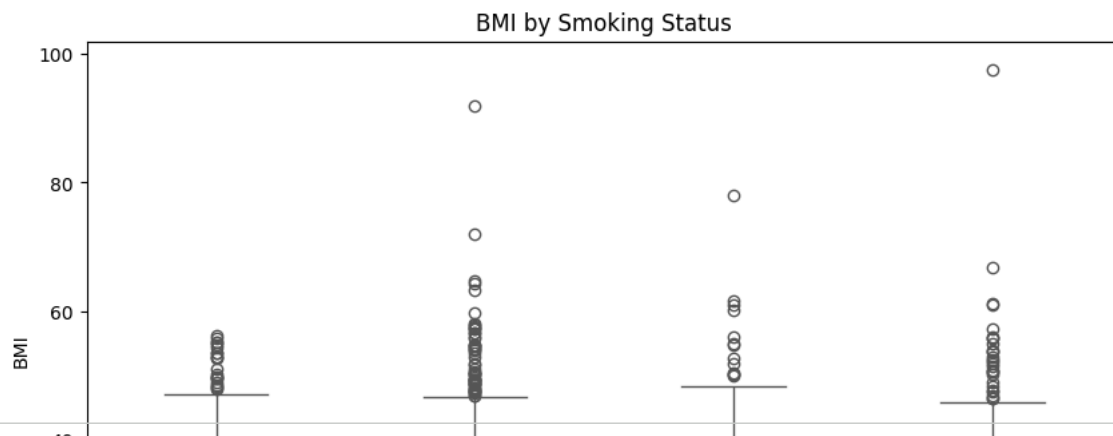


```
#Scatter plot: BMI VS average glucose level
plt.figure(figsize=(5,4))
sns.scatterplot(x='bmi', y='avg_glucose_level', data=df,hue='gender')
plt.title('BMI vs Average Glucose Level')
plt.xlabel('BMI')
plt.ylabel('Average Glucose Level')
```

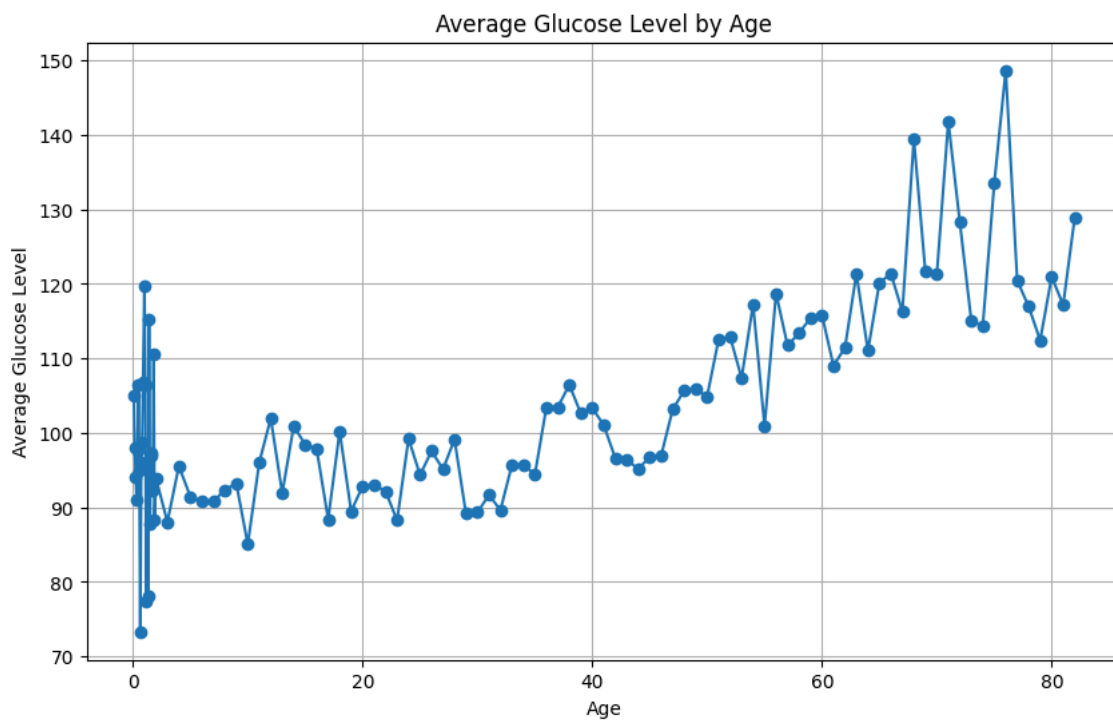
```
Text(0, 0.5, 'Average Glucose Level')
```



```
#Box plot:-BMI by Smoking Status
plt.figure(figsize=(10,6))
sns.boxplot(x='smoking_status', y='bmi', data=df,hue='smoking_status',palette='Set2',legend=False)
plt.xticks(rotation=45)
plt.title('BMI by Smoking Status')
plt.xlabel('Smoking Status')
plt.ylabel('BMI')
plt.show()
```



```
#Line Plot: Average Glucose Level by Age
avg_glucose_by_age = df.groupby('age')['avg_glucose_level'].mean()
plt.figure(figsize=(10,6))
plt.plot(avg_glucose_by_age.index, avg_glucose_by_age.values, marker='o', linestyle='--')
plt.title('Average Glucose Level by Age')
plt.xlabel('Age')
plt.ylabel('Average Glucose Level')
plt.grid(True)
plt.show()
```



```
#Pairplot
sns.pairplot(df,vars=['age','avg_glucose_level', 'bmi'],hue='gender')
plt.show()
```

